BUS 411 — Fixed Income Security Analysis and Valuation
Beedie School of Business
Exam 1 - June 23, 2026

Name: ______________________________

Student ID: ______________________________

Instructions

- This exam is closed book.
- You may bring one single-sided letter-size cheat sheet of notes and formulas. It must be submitted with your exam.
- Financial calculators are allowed.
- Show all work for numerical problems. Unsupported final answers receive limited credit.
- State assumptions clearly when a question requires judgment.
- In Section A, answer *only* 4 of the 5 short-answer questions. Each answered question is worth 10 points.
- In Section B, answer *only* 3 of the 4 problems. Each answered problem is worth 20 points.

Good luck.

A. Short Answer Questions

Question A1

A pension fund is choosing between a newly issued 8-year fixed-rate corporate bond, an 8-year floating-rate note indexed to SOFR, and an 8-year callable agency bond.

Explain how the main risk exposures differ across the three securities. Your answer should address interest-rate risk, reinvestment risk, credit or spread risk, and embedded-option risk. Also explain why a security with more stable price behavior can still be risky for the investor.

Question A2

Explain why yield to maturity is not the same thing as the realized return earned by a bond investor. In your answer, identify at least three assumptions embedded in yield to maturity and give one example where a bond with a higher yield to maturity may produce a lower realized return than another bond.

Question A3

A manager says: "My portfolio has a duration of 5.4, so I understand its interest-rate risk."

Critique this statement. Explain what duration captures, what it misses, and why convexity and key rate duration can change the risk assessment. Use the intuition of a nonparallel yield-curve shift in your answer.

Question A4

Investment-grade corporate bonds usually trade at positive spreads over Treasury securities with similar maturity.

Explain the economic reasons for this spread. Your answer should distinguish compensation for expected default loss, downgrade risk, liquidity risk, taxes or regulation when relevant, and market risk premia. Then explain why the observed spread is not a pure forecast of default losses.

Question A5

A bond portfolio manager is benchmarked to a broad bond index with thousands of securities. The manager is considering either a cell-based construction approach or a multi-factor model approach.

Compare the two approaches. Explain how each approach controls benchmark-relative risk, why pure bond index replication is difficult, and why tracking error rather than total portfolio volatility is the central risk measure for this manager.

B. Problems

Problem B1

A 6-year corporate bond has face value \$1,000, a 5.40% annual coupon paid semiannually, and a yield to maturity of 5.90% with semiannual compounding. Settlement occurs 74 days after the last coupon date in a 182-day coupon period. The matched 6-year Treasury yield is 4.35%.

Answer the following:

a. Compute the clean price of the bond per \$1,000 face value.

b. Compute accrued interest and the full price.

c. Compute the bond's yield spread over the matched Treasury yield.

d. Suppose the bond has modified duration of 5.05 and convexity of 31.8. Approximate the percentage price change if the bond's yield rises by 75 basis points.

e. Explain why the duration-convexity approximation may be less reliable for this corporate bond than for an option-free Treasury bond.

Problem B2

The following annual-pay Treasury securities are observed in the market. Assume all rates are annual effective rates and face value is \$100.

Maturity	Coupon rate	Price
1 year	0.00%	96.1538
2 years	5.00%	100.20
3 years	6.00%	101.10

Answer the following:

a. Compute the 1-year spot rate.

b. Bootstrap the 2-year spot rate.

c. Bootstrap the 3-year spot rate.

d. Compute the 1-year forward rate from year 2 to year 3.

e. Use the spot curve plus a constant 120 basis point spread to price a 3-year annual-pay corporate bond with a 5.50% coupon and face value \$100. Briefly explain what the constant spread assumption means.

Problem B3

A portfolio has active returns of 12 bps, -8 bps, 5 bps, 22 bps, -18 bps, and 4 bps over six months.

Answer the following:

a. Compute the mean active return.

b. Compute the sample monthly tracking error.

c. Annualize the tracking error.

d. Explain whether this is backward-looking or forward-looking tracking error.

e. Explain why this measure may be misleading if the manager changed the portfolio today.

Problem B4

A firm has issued a floating-rate note with quarterly coupons. The coupon rate each quarter is 3-month SOFR plus a quoted margin of 1.25%. The note has face value $5,000,000. There is a coupon floor of 2.00% and a cap of 6.50%. The next three SOFR reset rates are assumed to be:

Quarter	SOFR at reset
1	0.50%
2	3.80%
3	6.10%

Answer the following:

a. Compute the uncapped annual coupon rate for each quarter.

b. Apply the floor and cap.

c. Compute each quarterly coupon payment.

d. Explain how the cap changes investor exposure.

e. Explain why discount margin is more natural than yield to maturity for this security.